

Power Budget for Light Sensor - Jacob Alger							
Team Number:	Team 201						
Project Name:	Garden Buddy						
Team Member Names:	Jacob Alger, Raunak Singh Chhabra, Isaiah Johnston, Rylee Wirt						
Version:	3						
A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage regulators, resistors, capacitors, or passive elements							
All Major Components	Component Name	Part Number	SupplyVoltageRange	#	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
	Curiosity Nano Board	PIC18F57Q43	1.8V to 5.5V	1	500	500	mA
	Photoresistor	NSL-5152	100V	1	0.5	0.5	mA
	Op-Amp	MCP6002-E/P	1.8V to 6V	1	300	300	mA
B. Assign each major component above to ONE power rail below. Try to minimize the number of different power rails in the design.							
+5V Power Rail	Component Name	Part Number	SupplyVoltageRange	#	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
	Curiosity Nano Board	PIC18F57Q43	1.8V to 5.5V	1	500	500	mA
	Photoresistor	NSL-5152	100V	1	0.5	0.5	mA
	Op-Amp	MCP6002-E/P	1.8V to 6V	1	300	300	mA
	Subtotal					800.5	mA
	Safety Margin					25%	
	Total Current Required on +5V Rail					1000.625	mA
c2. Regulator or Source Choice	5V regulator	MC7805CT-BP	7V to 35V	1	1500	1500	mA
	Total Remaining Current Available on +5V Rail					499.375	mA

+9V Power Rail	Component Name	Part Number	SupplyVoltageRange	#	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
	5V regulator	MC7805CT-BP	7V to 35V	1	1500	1500	mA
	Subtotal					1500	mA
	Safety Margin					25%	
	Total Current Required on +9V Rail					1875	mA
c1. Regulator or Source Choice	DC Power Jack	PJ-102AH	24V	1	5000	5000	mA
	Total Remaining Current Available on +9V Rail					3125	mA
C. For each power rail above, select a specific voltage regulator using the same process as for major component selection. Confirm that the Total Remaining Current Available on							
D. Select a specific external power source (wall supply or battery) for your system, and confirm that it can supply all of the regulators for all of the power rails simultaneously. If							
External Power Source 1	Component Name	Part Number	SupplyVoltageRange	Output Voltage	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
Power Source 1 Selection	Plug-in Wall Supply	B09ZTKTLGW	110VAC	9V	15000	15000	mA
Power Rails Connected to	5V regulator	MC7805CT-BP	7V to 35V	5V	1500	1500	mA
External Power Source 1	DC Power Jack	PJ-102AH	24V	9V	5000	5000	mA
	Total Remaining Current Available on External Power Source 1					8500	mA